

UrbanRuleBench: Benchmarking Urban Intelligence Agents for GIS-Grounded Compliance Reasoning

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ABSTRACT

Large language models and agentic AI are increasingly used for urban planning support, geospatial analysis, and regulatory interpretation. Yet current evaluations rarely test the setting that matters for many planning decisions: whether an agent can connect an official statutory rule to a real parcel, interpret the relevant planning constraints, and return a machine-checkable compliance judgment rather than a plausible explanation. Existing benchmarks emphasize planning knowledge, simulated urban decision making, or general geospatial tool use, leaving parcel-specific regulatory compliance underexplored.

We present UrbanRuleBench, a Shenzhen-based benchmark for regulation-grounded and GIS-grounded urban compliance reasoning. The current release is organized as a two-layer evidence chain. The first layer, Source-419, contains 419 official statutory planning cases and evaluates structured extraction of rule fields such as plot identifiers, land-use codes, floor area ratio (FAR), land area, height, and facilities. The second layer, Parcel-GIS Direct, audits the subset of source cases that can be linked to real parcel geometry, producing 14 proposal-label cases over 7 parcels for direct compliance evaluation. This design separates the broad statutory-rule grounding problem from the stricter parcel-level GIS compliance claim, making the scope of evidence explicit.

Experiments with open-source LLMs show that high output validity does not imply reliable rule grounding: the best source-level rule-field F1 is only 26.36%, despite near-perfect schema validity for several models. On the parcel-GIS pilot subset, models can often match the final compliance label, but qualitative error analysis reveals brittle legal semantics, especially when models misread upper-bound constraints such as FAR limits as exact requirements. UrbanRuleBench therefore provides a reproducible benchmark interface, diagnostic metrics, and real Shenzhen case studies for evaluating urban intelligence agents at the boundary between statutory text, GIS evidence, and machine-verifiable compliance reasoning.

Keywords: Agentic AI, Urban Intelligence, GIS-Grounded Reasoning, Compliance Reasoning, Benchmark

INTRODUCTION

Artificial intelligence is moving from analytical support toward more active roles in urban planning, including document analysis, scenario generation, spatial analysis, and plan evaluation (Sanchez et al., 2023; Peng et al., 2024; Fu et al., 2025; Zheng et al., 2025b). This shift is consequential because planning decisions combine heterogeneous evidence: policy and statutory text, maps, parcel records, development controls, and site conditions. Large language models (LLMs) can synthesize these materials fluently, but fluency is not evidence of a correct planning judgment. Near statutory decisions, an answer may sound plausible while misreading a land-use permission, floor area ratio (FAR) limit, height control, or facility obligation. Planning-support agents must therefore produce decisions that are not only readable, but also traceable to the governing rule and spatial object.

This requirement extends a longer tradition of planning support rather than replacing it. Planning support systems were developed to combine geographic information, models, visualization, and scenario exploration in forms that assist professional judgment (Klosterman, 1997; Geertman and Stillwell, 2004; Pettit et al., 2018). Later work emphasized that useful support depends on how instruments fit planning processes, governance arrangements, and user needs, not only on technical sophistication (Geertman and

UrbanRuleBench Study Area and Shenzhen Dataset Geography

419 source-level statutory cases summarized by area count, plus 7 audited real parcel-GIS plot locations

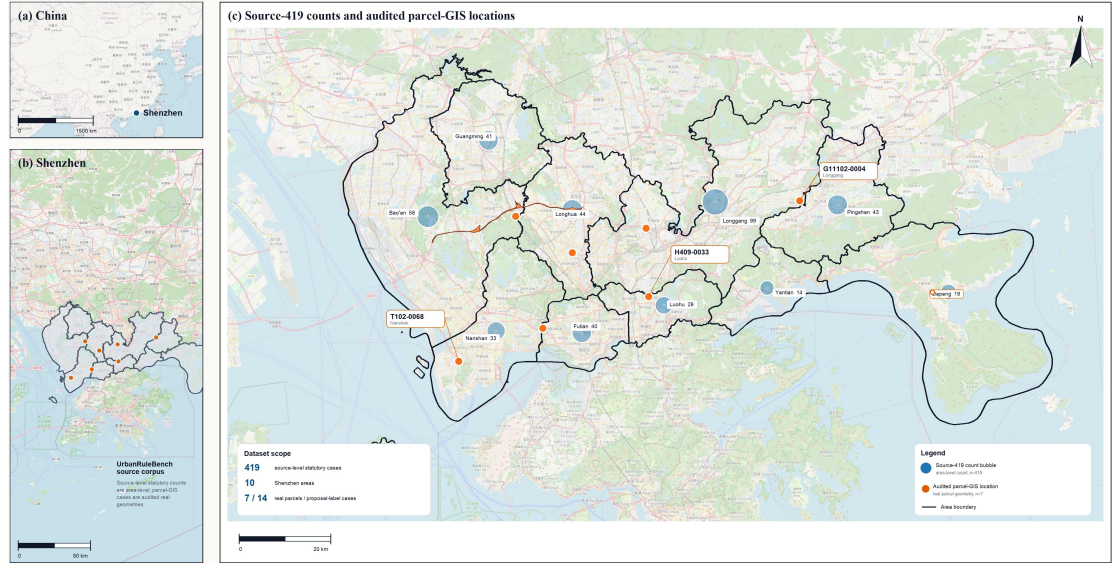


Figure 1. Study area and dataset geography of UrbanRuleBench in Shenzhen, China. Panel (a) locates Shenzhen within China, panel (b) shows the city-level spatial coverage of the benchmark, and panel (c) enlarges Shenzhen to show source-level coverage and audited parcel-GIS evidence. Blue proportional circles summarize the 419 Source-419 statutory rule cases by Shenzhen area; they are area-level counts, not parcel-level coordinates. Orange markers denote the 7 audited Parcel-GIS Direct parcels with real GIS geometry, which are used to construct 14 proposal-label compliance cases. The figure therefore separates broad statutory-source coverage from the stricter parcel-level GIS evidence used for direct compliance evaluation.

Stillwell, 2020). LLM-based agents inherit this socio-technical obligation while adding a new source of opacity: their explanations can be persuasive even when the underlying rule or spatial relation is wrong. Long-standing cautions about automating non-routine planning judgment and recent critiques of digital planning therefore make traceability and human review central design requirements (Batty, 2018; Potts et al., 2024).

This paper studies that requirement through *parcel-level compliance reasoning*. Given an official statutory planning source, a parcel, and a proposed development, an agent must recover the applicable controls, align them with the parcel and proposal, and return a machine-checkable compliance judgment. The task combines three forms of grounding. Regulatory grounding identifies the controlling rule fields and their operators; spatial grounding connects those fields to real parcel geometry; and decision grounding records the final label together with the violated dimensions. A benchmark that omits any one of these links cannot determine whether an agent reached the right answer from the right legal and spatial evidence.

Existing evaluations cover important parts of this problem but not the complete chain. Planning-focused benchmarks test professional knowledge and regulation-related questions, while architecture-focused evaluations include codes and planning knowledge in broader disciplinary assessments (Zheng et al., 2025a; Wu et al., 2025). Urban-agent benchmarks examine perception, spatiotemporal reasoning, simulation, and decision making across city tasks (Feng et al., 2024; Lai et al., 2025). Geospatial benchmarks evaluate spatial question answering, workflow generation, and multi-step GIS tool use (Xu et al., 2025; Krechetova and Kochedykov, 2025; Zhang et al., 2025a). Legal-language benchmarks, in turn, test legal understanding and several forms of legal reasoning without parcel geometry or planning-specific controls (Chalkidis et al., 2022; Guha et al., 2023). As summarized in Figure 2, these benchmarks reveal complementary capabilities, yet none directly evaluates whether an agent can move from an official statutory planning source to an audited real parcel and then adjudicate a proposal against typed development controls.

Automated compliance checking offers a closer conceptual precedent. This literature formalizes regulatory text as computable rules and links those rules to digital representations of buildings and their

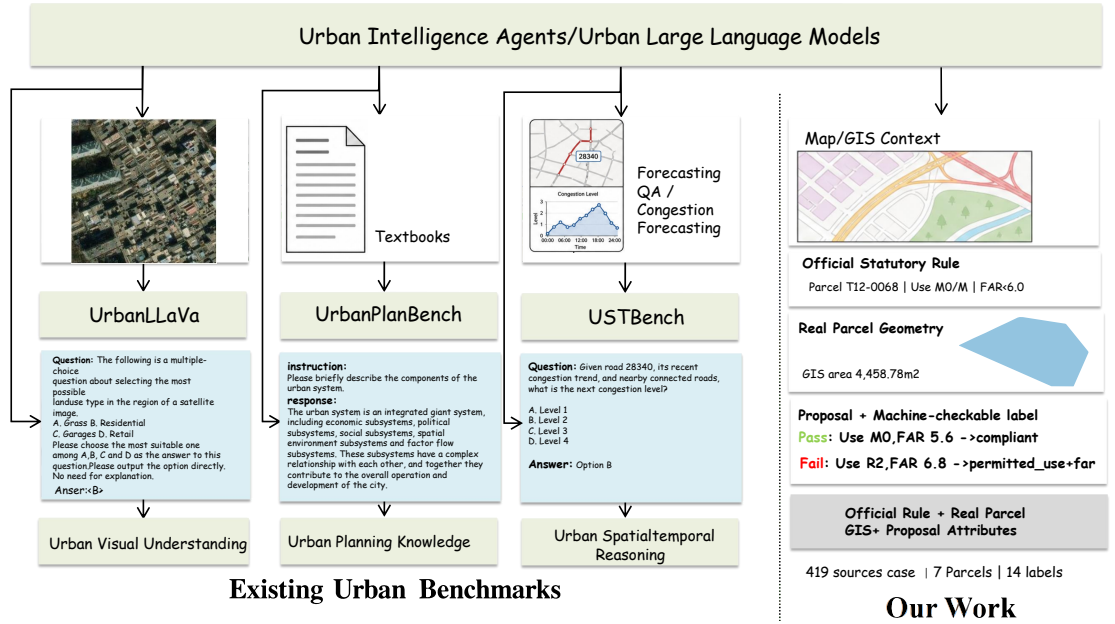


Figure 2. Positioning of UrbanRuleBench relative to existing urban intelligence benchmarks. Existing benchmarks commonly evaluate urban visual understanding, planning knowledge, or spatiotemporal reasoning from satellite imagery, textbook-style prompts, maps, and forecasting tasks. UrbanRuleBench instead evaluates the compliance setting in which an agent must combine map/GIS context, official statutory rules, real parcel geometry, and proposal attributes to produce machine-checkable pass/fail labels and violation dimensions. The figure therefore highlights the benchmark gap addressed by this paper: regulation-grounded and parcel-grounded compliance reasoning rather than general urban understanding alone.

surroundings (Eastman et al., 2009; Zhang and El-Gohary, 2016; Amor and Dimyadi, 2021). However, conventional compliance systems usually assume that rules and project data have already been represented in compatible forms. An LLM-based planning agent introduces an additional uncertainty: it must first recover the relevant controls from official text, preserve their semantics, and apply them to spatial evidence. Evaluating only the final label can conceal failures at any of these intermediate steps.

We introduce *UrbanRuleBench*, a Shenzhen-based benchmark that makes this evidence chain explicit. Its first layer, *Source-419*, contains 419 official statutory planning cases and evaluates structured extraction of plot identifiers, land-use codes and phrases, land area, FAR, height limits, and facility requirements. Its second layer, *Parcel-GIS Direct*, audits the subset that can be linked reliably to real parcel geometry and constructs 14 proposal-label cases over 7 parcels. Figure 1 distinguishes the broad statutory-source coverage from the smaller set of audited parcel-GIS cases. This two-layer design prevents source-level evidence from being mistaken for parcel-level grounding while preserving a reproducible path from official documents to machine-checkable judgments.

The paper makes three contributions. First, it presents a provenance-aware construction pipeline that converts official Shenzhen planning sources into structured rule-field instances and links only the auditable subset to parcel GIS evidence. Second, it introduces a layered evaluation protocol that separates output validity, statutory rule-field grounding, parcel-level compliance, and violation-dimension identification. Third, it provides empirical diagnostics across nine open-source LLMs: the best Source-419 rule-field F1 is 26.36% despite near-perfect schema validity for several models, and the parcel-GIS pilot exposes failures such as interpreting a maximum FAR as an exact requirement. Together, these contributions establish a cautious benchmark for testing whether urban agents can produce verifiable compliance judgments rather than merely plausible planning prose.

RELATED WORK

Planning support systems and accountable digital planning.

UrbanRuleBench belongs to a history of computational systems designed to support, rather than substitute for, planning judgment. Early planning support systems combined databases, spatial models, visualization, and scenario tools around specific professional tasks (Klosterman, 1997; Geertman and Stillwell, 2004). Their evolution toward smart-city and big-data settings broadened the available evidence and analytical scale (Pettit et al., 2018), but research continued to identify an implementation gap between technically capable prototypes and tools embedded effectively in planning practice (Geertman and Stillwell, 2020). This history matters for agentic AI. Planning decisions are institutional and distributive: they interpret rules, allocate development rights, mediate competing interests, and require public legitimacy. Scholars have consequently warned that AI is more credible for bounded support functions than for replacing non-routine planning judgment (Batty, 2018), and that digital planning can reproduce opacity, exclusion, and unequal power if technology is treated as neutral (Potts et al., 2024). Systematic reviews similarly call for human–AI collaboration and attention to the gap between experimental systems and practical governance (Son et al., 2023). UrbanRuleBench addresses one bounded but consequential support function. It does not automate planning approval; it tests whether an agent can expose the rule, parcel evidence, and violation logic on which a compliance recommendation would depend.

AI and language technologies for planning practice.

Planning research has long explored computational support for analyzing plans, policies, public input, and spatial scenarios. Recent reviews describe both the opportunities and institutional concerns surrounding AI-enabled planning, from decision support to increasingly automated plan-making (Sanchez et al., 2023; Peng et al., 2024). Within this broader agenda, natural language processing (NLP) has been used to read planning documents and recover policy themes (Fu et al., 2023); a field-level review nevertheless characterizes planning NLP as fragmented and calls for shared data, validation, and benchmarked evaluation (Fu, 2024). LLM studies extend this line by assessing plan quality and by outlining applications in policy synthesis, public participation, scenario generation, and design assistance (Fu et al., 2024, 2025; Zheng et al., 2025b). Domain benchmarks now test planning knowledge and architecture-related regulations: UrbanPlanBench evaluates planning principles, professional knowledge, management, and regulation questions, while Arch-Eval includes architectural regulations and urban planning within a broader professional knowledge system (Zheng et al., 2025a; Wu et al., 2025). These studies establish the need for domain-specific evaluation, but their knowledge- or document-centered tasks do not require an official rule to be linked to a real parcel and applied to a proposed development.

Urban agents and geospatial reasoning.

Urban-agent benchmarks move beyond document understanding toward interactive and spatially situated tasks. CityBench combines urban data and simulation to evaluate perception, understanding, and decision making across multiple cities (Feng et al., 2024); USTBench decomposes urban spatiotemporal reasoning into understanding, forecasting, planning, and reflection within an interactive environment (Lai et al., 2025). Geospatial evaluations test related but more operational capabilities. The benchmark of Xu et al. (2025) covers spatial understanding and route-planning tasks, GeoBenchX evaluates tool-calling agents on solvable and intentionally unsolvable multi-step geospatial problems (Krechetova and Kochedykov, 2025), and GeoAnalystBench assesses spatial-analysis workflows and code generation on expert-validated GIS tasks (Zhang et al., 2025a). Parallel work develops agents that plan and execute GIS workflows: GeoGPT connects natural-language requests to sequential geospatial tools (Zhang et al., 2024), while GTChain uses simulated tasks to train an open model to generate multi-step tool-use chains (Zhang et al., 2025b). At a more basic level, tests of vector geometries and topological predicates show that general LLMs remain inconsistent in formal spatial reasoning even when prompt engineering improves average performance (Ji et al., 2025). These studies expose failures in spatial relations, workflow construction, and long-horizon reasoning. UrbanRuleBench shares their emphasis on executable or verifiable outputs, but changes the object of evaluation: spatial operations are necessary evidence for parcel compliance, not the final task. The agent must additionally identify the governing rule, preserve its regulatory operator, and justify the decision against both statutory and parcel evidence.

Legal reasoning and evidence-augmented agents.

Legal NLP provides a second neighboring line because statutory planning requires specialized language understanding. LexGLUE standardizes multiple legal-language understanding tasks (Chalkidis et al., 2022), and LegalBench evaluates 162 tasks spanning several forms of legal reasoning (Guha et al., 2023). Chinese legal benchmarks sharpen the jurisdictional relevance of this problem: LawBench

evaluates legal knowledge across cognitive levels (Fei et al., 2024), while LexEval covers memorization, understanding, inference, discrimination, generation, and ethics across 23 tasks (Li et al., 2024). Work on compositional legal rules further shows that decomposing a rule into elements and recomposing their logical consequences can improve LLM reasoning (Servantez et al., 2024). These resources demonstrate that legal competence is multidimensional, but they do not model the GIS objects or development-intensity calculations needed for parcel adjudication. More generally, retrieval-augmented and tool-using methods allow language models to consult external evidence and execute actions (Lewis et al., 2020; Yao et al., 2023; Schick et al., 2023; Mialon et al., 2023). Such mechanisms are relevant to planning agents because official documents and GIS operations should remain external, inspectable evidence. Tool access alone, however, does not guarantee that the agent interprets a maximum threshold, permitted-use condition, or facility obligation correctly. UrbanRuleBench therefore evaluates the typed rule fields and violation dimensions that connect retrieved evidence to a compliance judgment.

Automated compliance checking and computable zoning rules.

Automated compliance checking provides the closest technical foundation. Early rule-checking systems represented design requirements explicitly and evaluated digital building models against them (Eastman et al., 2009); subsequent work classified rules by the information and computational operations required for execution (Solihin and Eastman, 2015). Reviews identify rule formalization, model quality, and data interoperability as persistent bottlenecks (Macit Ilal and Gunaydin, 2017; Amor and Dimyadi, 2021), while industry-facing research finds that organizational, commercial, and political barriers remain alongside technical ones (Beach et al., 2020). NLP methods have extracted regulatory requirements from natural-language documents (Zhang and El-Gohary, 2016); more recent approaches combine linguistic parsing with formal grammars for multi-requirement sentences (Zhou et al., 2022), or align regulatory concepts with Industry Foundation Classes representations using transformers (Zhang and El-Gohary, 2023). A systematic review of machine-learning-based code checking concludes that broad, highly automated coverage across regulations remains unresolved (Alnuzha and Bloch, 2025). Zoning compliance additionally requires context beyond a building model. BIM–GIS research therefore connects project information to parcels, roads, setbacks, and other neighborhood-scale objects referenced by zoning rules (Demir Altintas and Ilal, 2021, 2022). UrbanRuleBench differs in both scope and evaluation target. It does not propose a complete rule engine or assume that every source has a parcel-ready digital representation. Instead, Source-419 tests whether an LLM-based agent can recover typed controls from official statutory text, and Parcel-GIS Direct tests the smaller audited subset for which those controls can be applied to real geometry and proposal attributes. This explicit source-to-parcel boundary is the benchmark’s principal distinction from both general LLM evaluations and closed-form compliance systems.

METHODOLOGY

UrbanRuleBench is formulated as a benchmark for evaluating regulation-grounded and GIS-grounded compliance reasoning, not as a collection of isolated prompts. The central design principle is a source-to-parcel evidence chain. Official statutory planning sources are first represented as traceable source records, then converted into structured rule-field supervision, and only the auditable subset is linked to parcel geometry for proposal-level compliance evaluation. This separation is methodologically important: a statutory source can support rule-grounding evaluation without supporting a full parcel-level adjudication, whereas a valid parcel-GIS case requires explicit source-to-parcel alignment, reliable geometry, and machine-checkable rule labels. Figure 3 summarizes this benchmark construction logic and the evaluation protocol built around it.

Statutory source registry.

The source registry is built from official Shenzhen statutory planning materials rather than synthetic examples. Statutory planning documents function as institutional instruments that allocate development rights, specify land-use permissions, constrain spatial form, and impose public-facility obligations. For this reason, UrbanRuleBench treats provenance as part of the benchmark instance. The current release contains 419 statutory source cases across 10 administrative areas, including 229 statutory-plan result publications and 190 final local-adjustment publications. For each source, the registry records a source identifier, district, date, planning stage, document title, URL, local text path, attachment count, plan-area names, statutory plot identifiers, and any explicit parcel numbers or parcel codes detected in the document.

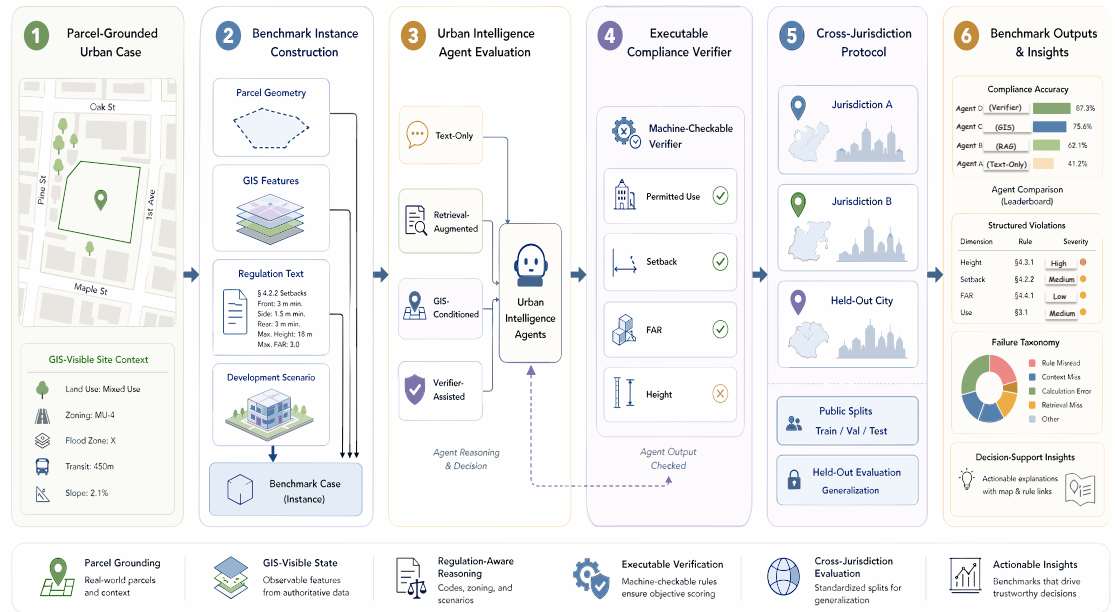


Figure 3. Overview of the UrbanRuleBench framework and full evaluation protocol. The benchmark is organized as a source-to-parcel evidence chain: official statutory planning sources are converted into structured rule-field cases, auditable parcel references are matched to GIS geometry, and proposal-label instances are evaluated with compliance metrics and violation-dimension diagnostics. The current release instantiates this framework through Source-419 statutory rule-field extraction and a 14-case Parcel-GIS Direct pilot subset. The text-only, retrieval-augmented, GIS-conditioned, verifier-assisted, and cross-jurisdiction components define the broader protocol for future benchmark expansion rather than fully completed experiments in this paper.

These fields make each case traceable to an official planning source and preserve the institutional context needed to distinguish district coverage, document type, and regulatory stage. This registry defines Source-419, the broad statutory-rule grounding layer of UrbanRuleBench.

Rule-field schema and source-level task.

Source-419 is represented as a structured extraction task over legally relevant planning controls. Given source metadata and evidence snippets from the official planning text, an agent must output a set-valued JSON object over seven rule fields: plot identifier, land-use code, land-use phrase, land area, floor area ratio (FAR), height limit, and facility phrase. These fields are selected because they correspond to controls that commonly determine whether a development proposal can be evaluated for approval. The plot identifier anchors a rule to a spatial planning unit; land-use code and land-use phrase represent permitted or intended use; land area and FAR define development intensity; height limits constrain the building envelope; and facility phrases capture public-service or supporting-facility obligations. The schema is intentionally source-level: it tests whether an agent can recover typed statutory controls before the benchmark asks whether a proposed development complies with those controls on a real parcel.

Claim boundary and parcel-GIS audit.

The transition from statutory source text to parcel-level GIS evidence is treated as an explicit audit problem. Many statutory planning documents define rules for plots, plan areas, or adjustment scopes, but do not provide enough information to identify a real parcel geometry with confidence. UrbanRuleBench therefore requires a Parcel-GIS Direct case to satisfy three evidentiary conditions: explicit source-to-parcel alignment, reliable parcel geometry, and auditable rule labels that can be checked against proposal attributes. Among the 419 source cases, 11 contain explicit parcel identifiers in the source registry, and 5 source documents support direct source-to-parcel alignment. After manual audit and geometry matching, these sources yield 7 real parcel references. This audit prevents the benchmark from treating every statutory source as a parcel-grounded GIS case and clarifies which claims are supported by source-level text versus direct parcel geometry.

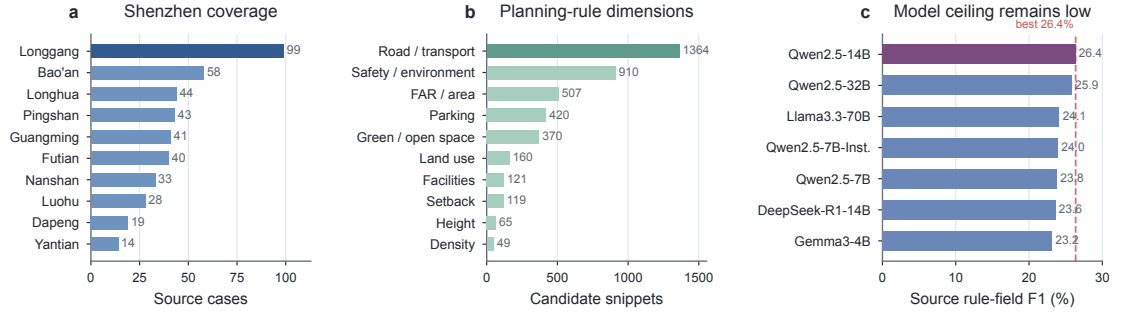


Figure 4. UrbanRuleBench dataset and diagnostic evidence. (a) Distribution of 419 statutory source cases across Shenzhen areas. (b) Coverage of planning-rule dimensions extracted from standards-rule candidate snippets. (c) Source-level rule-field F1 of valid models, reported as a diagnostic of statutory rule-grounding reliability rather than as a standalone model leaderboard.

Parcel-GIS Direct cases.

The audited parcel references are instantiated as the Parcel-GIS Direct split. From the audited parcel set, we construct 14 proposal-label cases by pairing each real parcel with compliant and noncompliant proposal scenarios whenever the rule state supports such a contrast. Each case combines official regulation text, source URL and title, structured rules, parcel GIS state, proposal attributes, expected compliance label, and violation objects when applicable. The structured rules include permitted land-use codes, maximum FAR, source rule area, parcel identifier, and audit status. The GIS state includes real parcel geometry, boundary path, centroid or bounding-box information, and visible urban context. Proposal attributes are evaluated against machine-checkable constraints such as permitted use, FAR upper bounds, height limits when available, facility requirements, and other typed violation dimensions represented in the case. Gold labels are produced through a deterministic compliance interface that checks proposal use, FAR, and available facility constraints, making each label auditable while preserving the limited scope of the pilot split.

Evaluation interface.

UrbanRuleBench exposes the two evidence layers through diagnostic evaluation metrics. Source-419 evaluates statutory rule grounding with schema validity and rule-field precision, recall, and micro-F1. Parcel-GIS Direct evaluates proposal-level compliance with JSON rate, valid rate, accuracy, balanced accuracy, noncompliant F1, and violation-dimension F1. These metrics separate formatting reliability from planning correctness: a model may produce valid structured output while omitting a legally relevant rule field, or it may match a final compliant/noncompliant label without identifying the violated regulatory dimension. We also define a broader protocol ladder—text-only agents, retrieval-augmented agents, GIS-conditioned agents, and verifier-assisted agents—to support future evaluations under increasing levels of evidence access and tool support. In the current release, the completed evidence consists of Source-419 and the Parcel-GIS Direct pilot; the benchmark should not be read as a completed cross-jurisdiction or city-scale approval engine.

EXPERIMENTS

Experimental Setup. The experiments are designed as diagnostic tests of urban intelligence agents rather than as a simple model leaderboard. They answer three questions. First, can current open-source LLMs recover structured statutory rule fields from official planning text? Second, when structured rules and real parcel GIS state are provided, can models produce machine-checkable compliance judgments? Third, do output validity and final-label accuracy reveal the same failures as field-level and violation-dimension metrics? We answer these questions with two connected splits: Source-419, a 419-case source-level statutory rule-grounding benchmark, and Parcel-GIS Direct, a deliberately scoped 14-case parcel-level pilot over 7 audited real parcels. The reported experiments instantiate the completed source-level statutory grounding task and the GIS-conditioned parcel-level pilot. Retrieval-augmented, verifier-assisted, and cross-jurisdiction variants are part of the broader UrbanRuleBench protocol for future expansion, not fully completed experiments in the current release.

Models and Prompting. All models are evaluated through the same JSON-oriented interface used by the benchmark scripts. The evaluated open-source model families include Qwen2.5 (Yang et al., 2024), Gemma 3 (Gemma Team, 2025), Llama 3.1 and Llama 3.3 (Dubey et al., 2024; Meta, 2024), DeepSeek-R1 (DeepSeek-AI et al., 2025), and QwQ-32B (Qwen Team, 2025). For Source-419, each model receives source metadata and evidence snippets and must output structured planning-rule fields. For Parcel-GIS Direct, each model receives official rule excerpts, structured rule state, parcel GIS state, and proposal attributes, and must output a JSON compliance decision with violation dimensions. This interface reflects the intended use case: a planning-support agent must return outputs that can be inspected, normalized, and checked by downstream tools or human planners. Low JSON Rate and low Valid Rate are therefore not treated as nuisance implementation details. They are reliability failures in their own right, because an unparseable or schema-invalid response cannot support a trustworthy statutory planning workflow even if its prose appears plausible.

Evaluation Metrics. The metrics intentionally separate formatting reliability from planning correctness. Valid Rate and JSON Rate evaluate whether a model can produce usable structured output. Rule-field precision, recall, and micro-F1 evaluate statutory rule grounding by comparing predicted value sets against gold source fields. Accuracy and Balanced Accuracy evaluate the final compliant/noncompliant label on Parcel-GIS Direct. Noncompliant F1 focuses on violation detection by treating noncompliance as the positive class, while Violation Dimension F1 evaluates whether the model identifies the correct regulatory reason for noncompliance, such as FAR, permitted use, or facility constraints. This layered design follows the logic of professional design and planning evaluation: a usable review must identify not only a final judgment, but also the evidence and rule dimension supporting it. Valid JSON is therefore not equivalent to legally meaningful compliance reasoning.

Let N be the number of evaluated cases, N_{json} the number of responses parseable as JSON, and N_{valid} the number of responses that satisfy the task schema after normalization. We define:

$$\text{JSON Rate} = \frac{N_{\text{json}}}{N}, \quad \text{Valid Rate} = \frac{N_{\text{valid}}}{N}. \quad (1)$$

For source-level rule grounding, each case is evaluated as a set-valued extraction problem. Let Y denote the gold rule-field values and $\hat{Y}$ denote the model-predicted values after normalization. Field-level precision, recall, and F1 are:

$$\begin{aligned} P &= \frac{|\hat{Y} \cap Y|}{|\hat{Y}|}, \\ R &= \frac{|\hat{Y} \cap Y|}{|Y|}, \\ F_1 &= \frac{2PR}{P+R}. \end{aligned} \quad (2)$$

For Parcel-GIS Direct, final-label performance is computed with standard binary compliance metrics:

$$\begin{aligned} \text{Acc} &= \frac{TP + TN}{TP + TN + FP + FN}, \\ \text{Bal. Acc.} &= \frac{1}{2} \left(\frac{TP}{TP + FN} + \frac{TN}{TN + FP} \right). \end{aligned} \quad (3)$$

Violation-dimension F1 applies the same F1 form to the set of predicted and gold violation dimensions, such as FAR, permitted use, and facility constraints:

$$F_1^{\text{dim}} = \frac{2P_{\text{dim}}R_{\text{dim}}}{P_{\text{dim}} + R_{\text{dim}}}. \quad (4)$$

For numerical planning controls, the verifier treats maximum FAR as an upper-bound constraint rather than an exact target:

$$\text{compliant}_{\text{FAR}} = \mathbb{I}(\text{FAR}_{\text{proposal}} \leq \text{FAR}_{\text{max}}). \quad (5)$$

Source-Level Statutory Rule Grounding. Table 1 shows that current open-source models can often produce parsable outputs but remain weak at statutory rule-field grounding. Qwen2.5-14B obtains the best rule-field F1 at 26.36%, driven by very high precision but low recall. Qwen2.5-32B reaches perfect schema validity and a similar F1 of 25.86%, suggesting that larger model size alone does not remove the recall bottleneck. The dominant failure mode is conservative under-extraction: models identify some obvious fields but miss many legally relevant controls. In zoning-code reasoning, these omissions are not harmless. Missing a FAR limit, height limit, land-area value, permitted-use code, or facility requirement can change the downstream compliance decision, even when the response is formatted as valid JSON.

Table 1. Main results on the 419-case source-level statutory rule benchmark.

Model	Valid Rate	Precision	Recall	Rule-field F1
Gemma3-4B	95.94%	43.86%	15.76%	23.18%
Qwen2.5-7B	99.05%	60.14%	14.87%	23.85%
Qwen2.5-7B-Instruct	99.52%	59.60%	15.02%	23.99%
Llama3.1-8B	15.27%	76.68%	1.00%	1.97%
Qwen2.5-14B	99.52%	92.69%	15.37%	26.36%
DeepSeek-R1-14B	96.42%	64.63%	14.45%	23.62%
Qwen2.5-32B	100.00%	90.44%	15.09%	25.86%
QwQ-32B	11.93%	88.65%	1.37%	2.69%
Llama3.3-70B	100.00%	56.78%	15.33%	24.14%

Table 2. Field-level performance on the source-level benchmark using Qwen2.5-14B.

Rule field	Precision	Recall	F1
Plot ID	94.82%	64.62%	76.86%
Land-use code	100.00%	12.03%	21.48%
Land-use phrase	81.31%	5.00%	9.43%
Land area	73.33%	4.76%	8.94%
FAR	29.55%	11.40%	16.46%
Height limit	72.73%	2.12%	4.11%
Facility phrase	66.67%	3.28%	6.25%

Table 2 explains why the aggregate F1 remains low. Plot identifiers are comparatively easy, reaching 76.86% F1 for Qwen2.5-14B, because they often appear in document titles, tables, or metadata. Numerical and semantic planning controls are much harder. FAR reaches only 16.46% F1, land area 8.94%, facility phrase 6.25%, and height limit 4.11%. This pattern is consistent with the structure of statutory planning documents: the named plot may be visible near the surface of the document, whereas intensity controls, land-use phrases, facility obligations, and dimensional limits may be embedded in dense text or tabular provisions and require normalization. Statutory rule grounding is therefore not solved by finding the parcel or plot name. The agent must recover typed planning controls that can support later compliance adjudication. This result supports the benchmark’s two-layer design: weak source-level grounding will propagate into unreliable parcel-level compliance unless rule extraction is explicitly evaluated.

Parcel-GIS Direct Compliance Evaluation. The Parcel-GIS Direct split evaluates 14 proposal-label cases over 7 audited parcels, with a balanced label design of 7 compliant and 7 noncompliant proposals. Each case is grounded in a direct textual parcel reference and real parcel geometry. The split is deliberately small by construction; its primary role is to test whether the source-to-parcel evidence chain can support verifiable compliance reasoning, not to establish a stable large-scale leaderboard. Table 3 shows that several models reach high final-label accuracy on this pilot, including perfect final-label accuracy with full structured validity for Qwen2.5-7B, Qwen2.5-7B-Instruct, and Qwen2.5-32B. These results should be read together with JSON Rate, Valid Rate, Noncompliant F1, and Violation Dimension F1. Models with low JSON Rate, such as Llama3.1-8B or QwQ-32B, should not be treated as reliable planning-support agents even when their conditional label scores look high. Conversely, high final-label accuracy does not

UrbanRuleBench Real Parcel Compliance Case

T102-0068 · Nanshan District, Danananshan Area, Shenzhen · real parcel geometry + statutory rule + EasyMap/OSM context + executable verifier

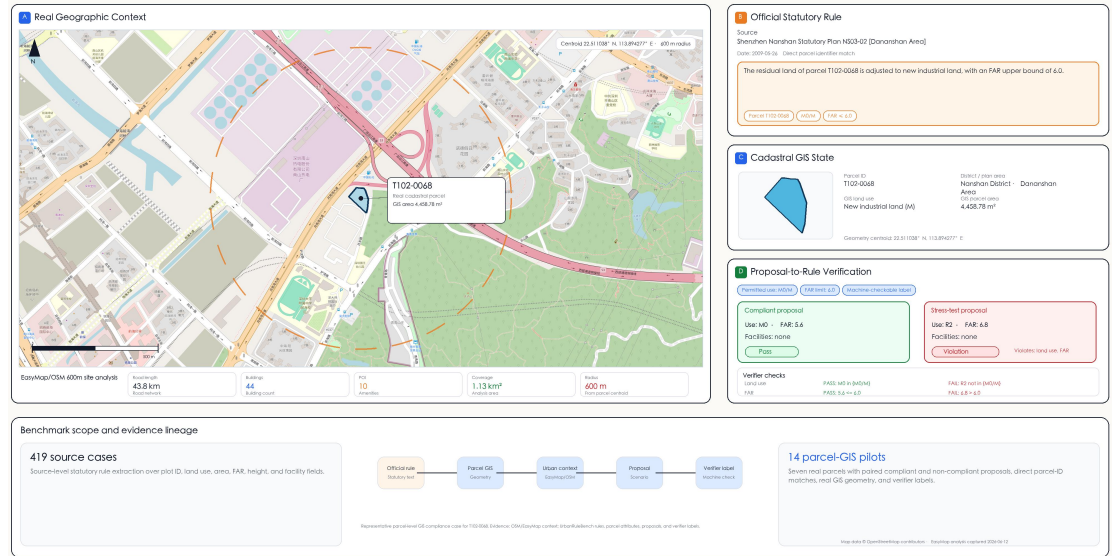


Figure 5. Example of a real parcel-level compliance case in UrbanRuleBench. The case is constructed around parcel T102-0068 in Danananshan Area, Nanshan District, Shenzhen, and integrates four types of evidence: real geographic context from EasyMap/OpenStreetMap, an official statutory rule excerpt, cadastral GIS state, and a proposal-to-rule verification module. The verifier compares proposed development scenarios against machine-checkable rule constraints, including permitted land use and floor area ratio limits, and returns structured compliance outcomes. The bottom evidence-lineage panel shows how source-level statutory rules are linked to parcel GIS geometry, urban context, proposed scenarios, and executable verifier labels, enabling objective evaluation of parcel-grounded compliance reasoning.

guarantee complete regulatory reasoning, because Violation Dimension F1 is lower and more variable than final-label accuracy for several models. Figure 5 illustrates the evidence chain for one real parcel.

Error Analysis and Regulatory Semantics. Figure 6 shows that even strong models can make semantically brittle compliance errors. The most important observed error is the misinterpretation of upper-bound controls, especially FAR limits, as exact requirements. This is a serious planning-semantics failure. In zoning and statutory planning, a maximum FAR defines an upper bound on development intensity; a proposal below the maximum is not automatically noncompliant merely because it does not equal the limit. Treating an upper bound as an exact target can therefore convert a compliant lower-intensity proposal into a false violation. Other possible failure modes include missing permitted-use constraints, overlooking facility requirements, confusing the presence of a parcel identifier with full rule understanding, and failing to identify the correct violation dimension. These errors are precisely why UrbanRuleBench reports violation-dimension metrics in addition to final-label accuracy.

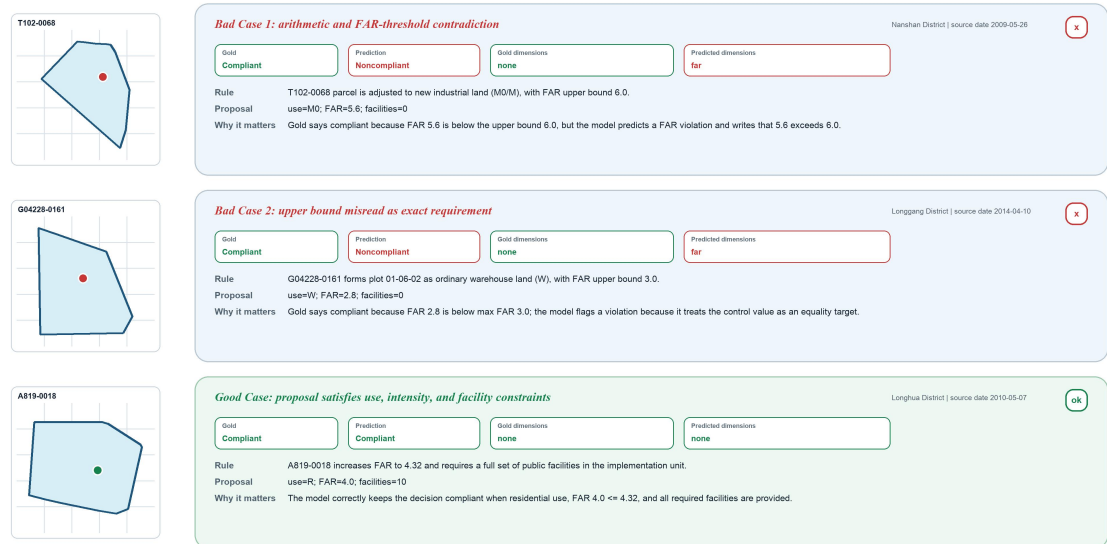


Figure 6. Good and bad parcel-level GIS compliance cases generated by Qwen2.5-14B on UrbanRuleBench. Each row shows a real parcel geometry, the gold compliance label, the model prediction, and the diagnostic reason. The bad cases show that valid JSON output does not guarantee reliable planning-rule semantics, especially when a model misreads FAR upper bounds as exact development requirements.

Table 3. Results on the 14-case Parcel-GIS Direct pilot subset. The split is reported as diagnostic evidence rather than a stable leaderboard. All evaluated models are retained for consistency with Table 1; Valid Rate and JSON Rate should be read before label metrics because some models return few usable structured predictions.

Model	Valid Rate	JSON Rate	Accuracy	Balanced Acc.	Noncompliant F1	Violation Dim. F1
Gemma3-4B	100.00%	100.00%	71.43%	71.43%	77.78%	58.33%
Qwen2.5-7B	100.00%	100.00%	<u>100.00%</u>	<u>100.00%</u>	<u>100.00%</u>	70.00%
Qwen2.5-7B-Instruct	100.00%	100.00%	<u>100.00%</u>	<u>100.00%</u>	<u>100.00%</u>	70.00%
Llama3.1-8B	92.86%	7.14%	<u>100.00%</u>	<u>100.00%</u>	<u>100.00%</u>	14.29%
Qwen2.5-14B	100.00%	100.00%	78.57%	78.57%	82.35%	81.48%
DeepSeek-R1-14B	42.86%	14.29%	83.33%	75.00%	66.67%	50.00%
Qwen2.5-32B	100.00%	100.00%	<u>100.00%</u>	<u>100.00%</u>	<u>100.00%</u>	<u>96.00%</u>
QwQ-32B	85.71%	0.00%	66.67%	60.00%	33.33%	0.00%
Llama3.3-70B	100.00%	92.86%	92.86%	92.86%	93.33%	72.73%

Evidence Status and Limitations. The experimental evidence supports two claims with different strengths. The 419-case Source-419 benchmark provides broad evidence that statutory rule-field grounding remains difficult for current open-source models. The 14-case Parcel-GIS Direct split provides pilot evidence that the source-to-parcel evidence chain can support real parcel-level compliance evaluation with machine-checkable labels. The main limitation is the availability of audited source-to-parcel matches with reliable geometry and labels. Future experiments should expand the parcel-GIS layer and instantiate retrieval-augmented, verifier-assisted, and cross-jurisdiction evaluations after comparable source-to-parcel data are available.

DISCUSSION

Fluent and valid outputs are not enough. UrbanRuleBench exposes a critical failure mode of current urban intelligence agents: high linguistic fluency, valid JSON, or even a correct final label can coexist with weak statutory rule grounding and incorrect regulatory semantics. The Source-419 results show that models often under-extract rule fields from official planning text, while the Parcel-GIS Direct examples show that models may produce formally usable outputs while misunderstanding the rule that justifies the

decision. This is an important distinction for benchmark design. A planning-support system is not reliable simply because it returns a well-formed response; it must recover the relevant statutory controls and apply them correctly to the parcel and proposal under review.

Planning significance. This failure mode matters because planning rules are not ordinary text to summarize. Statutory planning documents are institutional controls over land use, development intensity, spatial form, public facilities, and development rights. A missed land-use control, FAR limit, height limit, or facility requirement can change what a planner, applicant, or public reviewer believes is permissible on a parcel. In this sense, UrbanRuleBench evaluates a form of planning reasoning that sits between document interpretation and site-specific judgment. The benchmark does not ask whether an agent can write plausible planning commentary; it asks whether the agent can recover the controls that structure development rights and connect them to real parcel evidence.

Regulatory semantics and zoning-code reasoning. The error analysis shows that compliance reasoning requires typed statutory semantics, not only retrieval or spatial reasoning. A salient example is the misreading of FAR upper bounds as exact requirements. In zoning and statutory planning, a maximum FAR constrains the upper limit of development intensity; a proposal below that limit is not automatically noncompliant. Similar failures can occur when a model misses permitted-use constraints, overlooks facility requirements, treats a detected parcel identifier as sufficient rule understanding, or predicts a final label without identifying the correct violation dimension. These errors are planning-semantics errors: the model may have seen the relevant words or parcel reference, but it has not represented the regulatory operator, threshold, or typed constraint needed for compliance adjudication.

Evidence boundary as methodology. The two-layer design is therefore a methodological contribution rather than a dataset inconvenience. Source-419 supports broad evaluation of statutory rule grounding across 419 official Shenzhen source cases. Parcel-GIS Direct supports a deliberately scoped real-parcel compliance pilot over the subset where source-to-parcel alignment, geometry, proposal attributes, and rule labels can be audited. Keeping these layers separate prevents the paper from overstating parcel-level GIS grounding when only statutory source text is available. It also creates a reproducible expansion path: future work can add more parcel-level cases by auditing additional source-to-parcel matches rather than by assuming that every statutory source already supports parcel adjudication.

Implications for trustworthy urban AI. For urban AI systems intended to support planning workflows, evaluation should be diagnostic rather than limited to final-answer accuracy. UrbanRuleBench separates output validity, statutory rule-field grounding, parcel/GIS grounding, violation-dimension identification, and final compliance labels. This layered view makes different risks visible: a model may fail to produce usable structured output, omit legally relevant controls, misread a zoning threshold, or reach the correct final label for the wrong reason. Such diagnostics are important before deployment in planning-support settings, where planners need to inspect not only what the agent decided but also which rule fields and spatial evidence supported the decision.

Benchmark design as risk decomposition. The layered metrics also connect UrbanRuleBench to a broader shift from single-score evaluation toward diagnostic testing. Behavioral evaluation has shown that aggregate accuracy can conceal systematic capability failures (Ribeiro et al., 2020), while holistic language-model evaluation argues for reporting multiple scenarios and metrics so that trade-offs remain visible (Liang et al., 2023). In planning compliance, this principle has a concrete institutional meaning. JSON Rate and Valid Rate indicate whether an output can enter a downstream workflow; rule-field F1 tests whether the governing controls were recovered; violation-dimension F1 tests whether the stated reason matches the rule; and final-label metrics test the adjudication outcome. No single measure substitutes for the others. The separation also supports human oversight: a planner can reject an ungrounded decision even when its binary label is correct. This is consistent with evidence that automated compliance checking is more acceptable when expert review is retained (Beach et al., 2020) and with warnings that digital planning systems should not obscure how technical choices distribute authority and risk (Potts et al., 2024).

Implications for planning workflow design. The diagnostic structure suggests how regulation-aware agents could be situated within a planning-support workflow without being mistaken for autonomous approval systems. An operational sequence would preserve distinct checkpoints: retrieve the authoritative

source, normalize the applicable controls, verify the source-to-parcel match, compare proposal attributes with typed constraints, and present the resulting label and violation dimensions for professional review. Each checkpoint exposes a different failure surface and can be inspected independently. This staged interpretation follows the planning-support principle that computational tools should fit an institutional task and strengthen human judgment (Klosterman, 1997; Geertman and Stillwell, 2020). It also reflects automated-compliance research showing that rule interpretation, model preparation, execution, and reporting are separate technical and organizational problems (Amor and Dimyadi, 2021; Beach et al., 2020). UrbanRuleBench does not implement or validate a complete municipal workflow, but its two evidence layers make these interfaces explicit enough to evaluate components progressively. Future deployment studies should therefore measure not only model accuracy, but also correction effort, reviewer agreement, provenance completeness, and the ability of planners to identify and override unsupported recommendations.

Limitations and future work. UrbanRuleBench should not be read as an automated planning approval system or as a complete city-scale compliance engine. The current release provides broad evidence for statutory rule-grounding difficulty through Source-419 and pilot evidence for parcel-level GIS compliance through 14 Parcel-GIS Direct cases over 7 audited real parcels. The main limitation is the difficulty of obtaining reliable source-to-parcel alignment, parcel geometry, and auditable labels at scale. Future work should expand the parcel-GIS layer, add more rule dimensions, instantiate retrieval-augmented and verifier-assisted evaluations, and test cross-jurisdiction transfer once comparable official sources and parcel geometries are available. These steps would extend UrbanRuleBench from a cautious Shenzhen-based benchmark into broader infrastructure for regulation-aware urban AI evaluation.

CONCLUSION

UrbanRuleBench evaluates whether urban intelligence agents can connect official statutory planning rules to real parcel evidence and return machine-checkable compliance judgments. Its two-layer design separates broad source-level rule grounding across 419 Shenzhen cases from direct parcel-level evaluation on 14 proposal-label cases over 7 audited parcels. This separation makes the evidentiary scope visible instead of treating every planning document as if it already supported a parcel-grounded compliance claim.

The experiments reveal a gap between output form and regulatory substance. Several evaluated open-source LLMs achieve near-perfect schema validity, yet the best Source-419 rule-field F1 is only 26.36%. The parcel-GIS pilot further shows that a model may return a valid structure or even the correct final label while misidentifying the governing violation dimension or misreading an upper bound such as maximum FAR. Reliable planning-support evaluation must therefore inspect the recovered controls, their legal semantics, the spatial evidence, and the final judgment together.

The present benchmark remains deliberately bounded. Parcel-GIS Direct is a small Shenzhen pilot, and the available evidence does not yet support claims of city-scale, cross-jurisdictional, or deployment-ready compliance automation. Expanding audited source-to-parcel matches is the immediate priority; subsequent work should add rule dimensions, retrieval-augmented and verifier-assisted settings, and comparable cases from other jurisdictions. Within these limits, UrbanRuleBench provides a reproducible foundation for evaluating regulation-aware urban agents by what they can substantiate, not merely by what they can state.

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